Kitchen Food Safety and Sanitation

My reviewer for our Home Economics specialization. Hotel kitchen, carinderia or home, the rule is the same: tasty food is useless if it makes people sick.

Three kinds of food hazards

A food hazard is anything in food that can cause illness or injury to the person eating it.

- Biological hazards: living things or things made by them, such as bacteria, viruses, parasites and molds. Salmonella in undercooked chicken and the hepatitis A virus spread by unwashed hands are examples. These cause most food poisoning cases.
- Chemical hazards: harmful substances that get into food. Examples are dishwashing liquid left on plates, pesticide residue on unwashed vegetables, or insect spray used near open food.
- Physical hazards: hard or sharp foreign objects. Examples are hair, glass from a broken jar, small bones, staples from a sack, or a piece of steel wool.

The temperature danger zone

Bacteria multiply fastest between 5 and 60 degrees Celsius. This range is called the temperature danger zone. In our hot climate, food left on the table for hours after a party can quickly become unsafe.

- Keep cold food cold, at 5 degrees Celsius or below.
- Keep hot food hot, at 60 degrees Celsius or above.
- Do not leave cooked food at room temperature for more than about 2 hours. Pack leftovers and refrigerate them promptly.

Cross-contamination

Cross-contamination is when harmful germs move from one food, surface or person to another food. A common example: slicing raw chicken on a board, then using the same unwashed board and knife to slice tomatoes for a salad that will not be cooked.

Ways to prevent it:

- Use separate boards and knives for raw and ready-to-eat foods.
- Store raw meat on the lowest shelf of the refrigerator so its juices cannot drip onto other food.
- Wash and sanitize tools and surfaces between tasks.

Color-coded chopping boards

A widely used color system:

- Red: raw meat
- Blue: raw fish and seafood
- Yellow: cooked meat
- Green: fruits and salad greens
- Brown: vegetables
- White: dairy and bread

Some kitchens follow a slightly different scheme, so always follow the chart posted in your workplace.

Proper handwashing

Hands are one of the biggest carriers of germs in a kitchen. Good handwashing goes like this:

- Wet your hands with clean running water.
- Apply soap and rub all parts, including the backs of your hands, between the fingers and under the nails, for at least 20 seconds.
- Rinse well under running water.
- Dry with a clean single-use paper towel or a clean dryer.

Wash your hands before handling food, and after using the toilet, touching raw meat, handling garbage, sneezing, or touching your face, hair or phone.

Cleaning vs sanitizing

These are two separate steps, and the order matters.

- Cleaning removes visible dirt, grease and food bits using water and detergent.
- Sanitizing reduces the number of germs to a safe level, using heat (like very hot water) or an approved chemical sanitizer.

You must clean first, because sanitizer cannot work properly on a surface still covered with grease and food scraps.

FIFO storage

FIFO means First In, First Out. Older stock is used before newer stock. When new deliveries of canned goods arrive, place them behind the older cans so the older ones get used first. Label containers with the date received or prepared. This reduces spoilage and waste.

Safe thawing methods

Never thaw frozen meat by leaving it out on the counter, because the outside enters the danger zone while the center is still frozen. Safe methods are:

- In the refrigerator, which is the safest but slowest, so plan a day ahead.
- Under cold running water, or submerged in cold water changed often, with the food sealed in a leak-proof bag.
- In the microwave, but only if you cook the food right away.
- As part of cooking, for small items like frozen dumplings or thin patties.

Personal hygiene in food service

- Take a bath daily and wear a clean uniform and apron.
- Tie your hair and wear a hairnet or cap.
- Keep fingernails short, clean and unpolished. No fake nails.
- Remove rings, bracelets and watches, since they trap germs and can fall into food.
- Cover cuts or wounds with a waterproof bandage, and wear gloves over it.
- Do not work with food when you have diarrhea, vomiting or fever. Tell your supervisor.
- No eating, smoking or chewing gum in the food preparation area.

- Cough or sneeze into a tissue or your elbow, away from food, then wash your hands.

Quick check

- Q: What is the temperature danger zone? A: 5 to 60 degrees Celsius, where bacteria grow fastest.
- Q: A piece of glass in a soup is what type of hazard? A: A physical hazard.
- Q: Which comes first, cleaning or sanitizing? A: Cleaning, because sanitizer does not work well on dirty surfaces.
- Q: What does FIFO mean? A: First In, First Out, so older stock is used before newer stock.
- Q: Name one safe way to thaw frozen chicken. A: In the refrigerator, under cold running water in a sealed bag, or in the microwave if cooked right away.